

ECA5601 – 5G/ 6G COMMUNICATION

Exp 10 - 5G Network Slicing Environment Using MATLAB

Aim:

To analyze the performance of a 5G Network Slicing environment by studying the number of network slices, slice utilization, and end-to-end latency using MATLAB.

Procedure :

1. Initialize MATLAB and define the 5G network parameters such as bandwidth, latency, data rate, and number of users.
2. Create different network slices for services such as enhanced Mobile Broadband (eMBB), Ultra-Reliable Low-Latency Communication (URLLC), and massive IoT (mMTC).
3. Allocate network resources such as bandwidth and processing capacity to each slice according to its service requirements.
4. Simulate the network performance by varying traffic load and measuring parameters such as throughput, latency, and resource utilization.
5. Plot and analyze the results in MATLAB to compare the performance of different network slices and verify efficient resource allocation.

Objective – 1: Analyze the Number of Network Slices

Program :

```
clc;  
clear;  
close all;  
slices = {'eMBB','URLLC','mMTC','Private','IoT'};
```

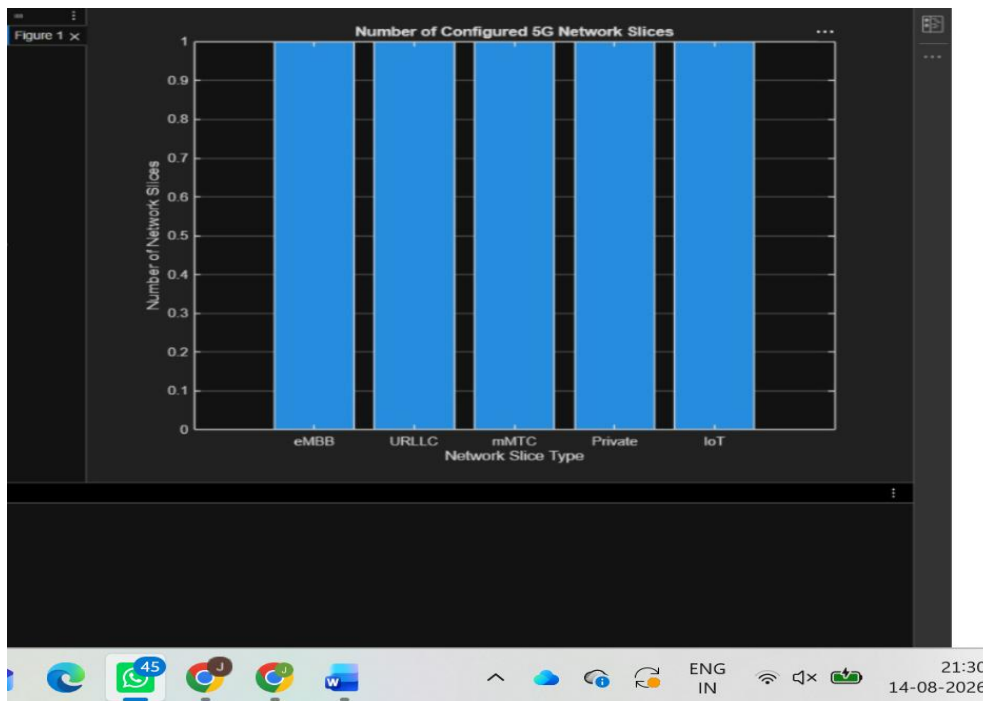
```

count = [1 1 1 1 1]; % One instance of each network slice

figure
bar(count)
grid on
set(gca,'XTickLabel',slices)
xlabel('Network Slice Type')
ylabel('Number of Network Slices')
title('Number of Configured 5G Network Slices')

```

Output :



Objective – 2: Compare Slice Utilization (%)

Program :

```

clc;
clear;
close all;

slices = categorical({'eMBB','URLLC','mMTC','Private','IoT'});
utilization = [85 70 60 75 50];

figure;

```

```

bar(slices, utilization);
grid on;
xlabel('Network Slice Type');
ylabel('Slice Utilization (%)');
title('Slice Utilization in 5G Network Slicing');
ylim([0 100]);

```

Output :



Objective – 3 : Analyze End-to-End Latency (ms)

Program :

```

clc;
clear;
close all;
slices = [1 2 3 4 5];
latency = [25 5 40 15 60];
figure;

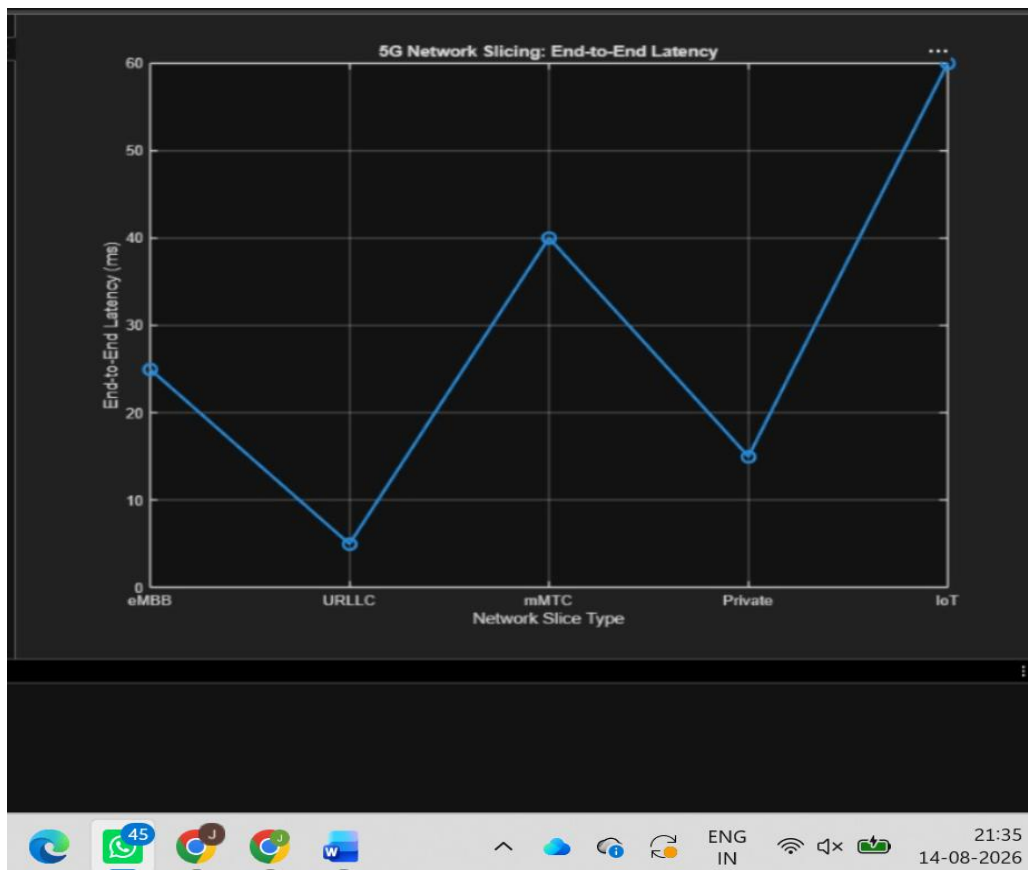
```

```

plot(slices, latency, '-o','LineWidth',2,'MarkerSize',8);
grid on;
xticks(slices);
xticklabels({'eMBB','URLLC','mMTC','Private','IoT'});
xlabel('Network Slice Type');
ylabel('End-to-End Latency (ms)');
title('5G Network Slicing: End-to-End Latency');
ylim([0 70]);

```

Output :



Result :

The MATLAB analysis successfully demonstrated the characteristics of a 5G Network Slicing environment. The results showed that multiple network slices can coexist while exhibiting different utilization levels and latency characteristics based on their intended services.